



PROJECT TITLE

For

Sensor Lab mini Project

**Third Year (Semester-VI) Bachelors in Information Technology, Autonomous
Program**

By

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Certificate

This is to certify that _____ have completed the project
report on the _____ topic _____ satisfactorily in
partial fulfilment of the requirements for the award of Mini Project in sensor
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Supervisor/ Examiner

Abstract

This project involves designing and implementing an **Obstacle Avoiding Car** using an **Arduino Uno, Motor Driver Shield, Ultrasonic Sensors, DC Motors, and a Servo Motor**. The car operates autonomously by detecting obstacles in its path and making real-time decisions to avoid collisions. This is achieved through the integration of **ultrasonic sensing, motor control, and decision-making algorithms** implemented using **Arduino programming**. The primary objective is to demonstrate the practical application of **sensor integration, embedded systems, and autonomous navigation**. This project serves as a fundamental example of how microcontrollers can be used to control robotic systems, with potential applications in areas such as **self-driving cars, industrial automation, and mobile robotics**.

Chapter 1: Introduction

An **Obstacle Avoiding Car** is an autonomous robotic vehicle designed to navigate its environment without colliding with obstacles by using an **Arduino Uno, Motor Driver Shield, Ultrasonic Sensor, Servo Motor, and DC Motors**. The Arduino Uno acts as the central controller, processing distance data from the ultrasonic sensor, which emits sound waves and measures their reflection to detect nearby objects. Mounted on a servo motor, the ultrasonic sensor scans the area by rotating within a specified angle range, providing distance measurements in multiple directions. When an obstacle is detected within a certain threshold, the car stops, scans left and right, and decides the best direction to move based on the available clearance. The Motor Driver Shield controls the DC motors, enabling the car to move forward, turn, or reverse as required. This continuous monitoring and decision-making process allow the car to autonomously navigate around obstacles, making it suitable for various applications like autonomous robotic vehicles, smart wheelchairs, and educational robotics projects.

Chapter 2: Literature Review

Survey

Various obstacle-avoidance systems are used in robotics, involving sensors such as Ultrasonic, Infrared (IR), and LIDAR. Ultrasonic Sensors are preferred for their accuracy, cost-effectiveness, and ease of implementation. Common implementations involve differential steering mechanisms to avoid obstacles.

Research Gap

Most obstacle-avoidance systems are either too expensive or complex for educational purposes. This project aims to develop a cost-effective, efficient, and simple obstacle-avoiding car using basic components.

Related Work

1. "Design of Obstacle Avoidance Robot" by A. K. Sahu and A. K. Dwivedi (2020): This paper presents the design and implementation of an obstacle-avoiding robot using ultrasonic sensors and Arduino. The paper highlights the effectiveness of using ultrasonic sensors for real-time obstacle detection and avoidance.
 2. "Obstacle Avoidance Robotic Vehicle Using Ultrasonic Sensors" by P. Gupta and S. Sharma (2019): This paper discusses an efficient obstacle-avoiding robotic vehicle controlled by ultrasonic sensors and servo motors. The design focuses on improving response time and accuracy in avoiding obstacles.
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Chapter 3: Design Procedure

Problem Definition

The objective of this project is to design and develop a low-cost, efficient Obstacle Avoiding Car using an Arduino Uno and commonly available components like Ultrasonic Sensors, Servo Motor, DC Motors, and a Motor Driver Shield. This project aims to demonstrate fundamental robotic principles such as obstacle detection, path planning, and autonomous navigation using simple electronic components and Arduino programming.

3.2 Steps Involved

1. Component Selection and Procurement

- Selecting appropriate components required for building the obstacle-avoiding car, including:
 - Arduino Uno (Microcontroller for processing sensor data and controlling motors)
 - Motor Driver Shield (L298N or L293D) (For controlling DC motors with sufficient power and directional control)
 - Ultrasonic Sensor (HC-SR04) (For detecting obstacles by measuring the distance from objects using sound waves)
 - Servo Motor (To rotate the ultrasonic sensor and enable wider scanning angles)
 - DC Motors (For driving the wheels of the car)
 - Chassis and Wheels (To provide structural support and mobility)
 - Power Supply (Batteries or External Source) (To power the entire system)
- Procuring these components from reliable sources to ensure compatibility and proper functioning.

2. Circuit Assembly and Wiring

- Assembling the car by mounting all components onto the chassis.
- Connecting the DC Motors to the Motor Driver Shield, which is then connected to the Arduino Uno.
- Wiring the Ultrasonic Sensor to the Arduino's digital pins for transmitting and receiving signals.
- Connecting the Servo Motor to the Arduino's PWM pins for controlled rotation.

- Ensuring proper power supply connections to avoid short circuits and ensure stable operation.

3. Arduino Programming

- Writing a program using the Arduino IDE to control the entire system.
- Programming the Arduino to:
 - Continuously monitor the distance measured by the Ultrasonic Sensor.
 - Rotate the Servo Motor to scan the surroundings for obstacles.
 - Make decisions based on distance data, determining whether to move forward, stop, or turn left/right.
 - Control the DC Motors via the Motor Driver Shield to execute movement commands.
- Implementing algorithms for obstacle detection and decision-making to achieve smooth navigation.

4. Testing and Calibration

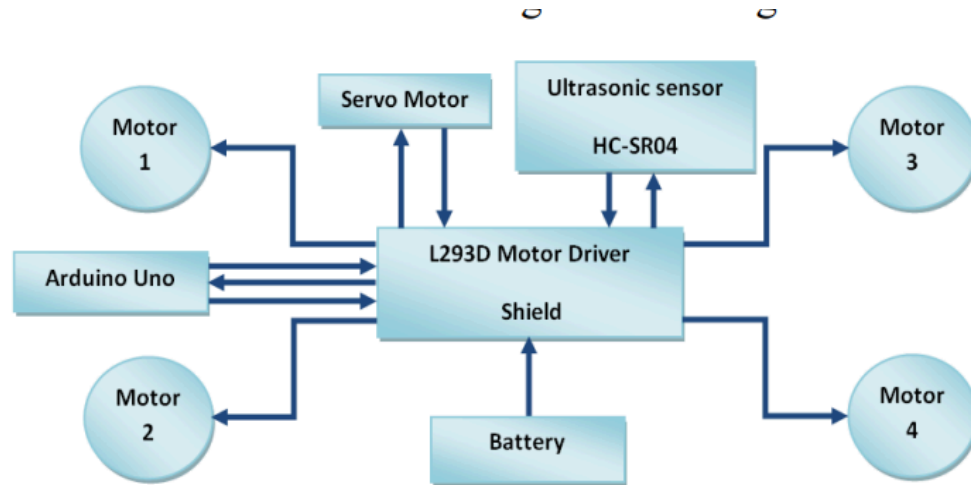
- Testing the car's functionality by running the programmed code and observing its response to obstacles.
- Calibrating the Ultrasonic Sensor for accurate distance measurement by adjusting threshold values in the code.
- Adjusting the Servo Motor angles for optimal scanning range.
- Fine-tuning motor control settings to ensure smooth turning and obstacle avoidance.

5. Final Assembly and Demonstration

- Securing all components properly onto the chassis after successful testing.
- Demonstrating the obstacle-avoiding capability by placing the car in various environments with obstacles.

- Evaluating the car's performance, stability, and efficiency in avoiding obstacles and navigating around them.

3.3 Block Diagram of Proposed Project



3.4 Component Description

Hardware Components

1. **Arduino Uno** - Controls the entire system and processes sensor data.
2. **Motor Driver Shield** - Mounted on Arduino, controls DC motors through PWM signals.
3. **Ultrasonic Sensor (HC-SR04)** - Measures distance to obstacles.
4. **Servo Motor** - Rotates ultrasonic sensor for directional scanning.
5. **DC Motors (4 Motors)** - Provide motion to the vehicle.
6. **Battery (9V/12V)** - Supplies power to the Arduino and motors.
7. **Switch** - Used to turn the system on or off.
8. **Connecting Wires** - Used for establishing connections.

Specifications

- **Arduino Uno:** 5V, 14 Digital I/O Pins, 6 Analog Inputs, Clock Speed: 16 MHz.
 - **HC-SR04:** Operating Voltage: 5V, Range: 2cm - 400cm, Accuracy: ± 3 mm.
 - **Motor Driver Shield:** Supports 2A per channel, 4 H-Bridges.
 - **Servo Motor:** Operating Voltage: 4.8V-6V, Angle Range: 0° - 180° .
 - **DC Motors:** 6V-12V, RPM: 100-300 (Depending on Motor Type).
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3.5 Working of Proposed Project

- The **Ultrasonic Sensor** continuously emits sound waves and measures the time taken for the echo to return.
 - The **Arduino Uno** calculates the distance to obstacles based on the speed of sound.
 - If an obstacle is detected within a threshold distance, the car stops and scans the environment using the **Servo Motor**.
 - The car decides the clear path and changes direction accordingly.
 - This process repeats continuously to achieve autonomous navigation.
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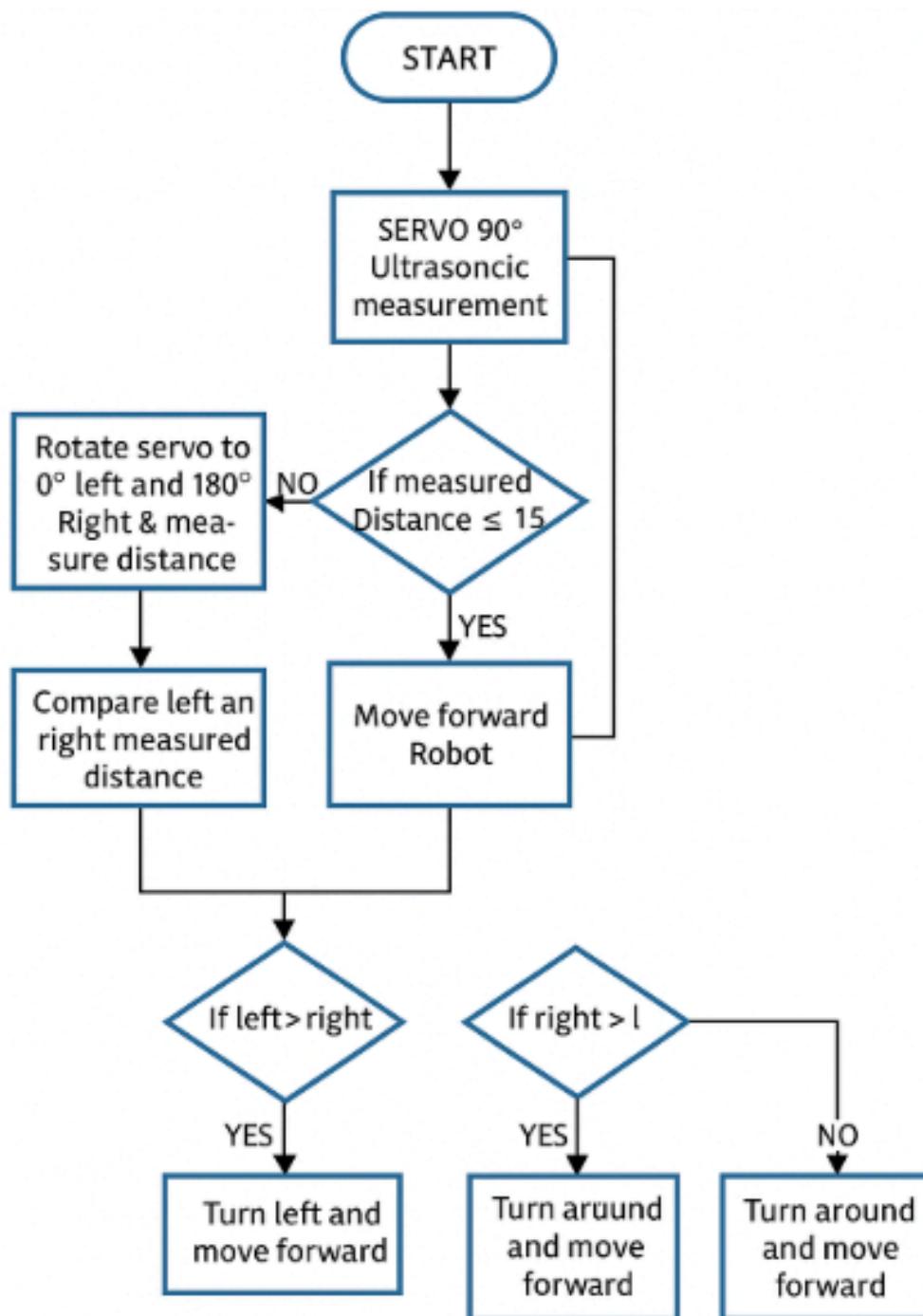
Chapter 4: Implementation

Hardware Implementation

- Connections between **Arduino, Motor Driver Shield, Ultrasonic Sensor, Servo Motor, and DC Motors**.
- Circuit diagram and wiring details.

Software Implementation

Flowchart :



- Arduino programming using C/C++ (Arduino IDE).

Code

```
//including the libraries
#include <AFMotor.h>
#include <NewPing.h>
#include <Servo.h>

//defining pins and variables
#define TRIG_PIN A4
#define ECHO_PIN A5
#define MAX_DISTANCE 200
#define MAX_SPEED 200 // sets speed of DC motors
#define MAX_SPEED_OFFSET 20
#define turn_amount 500

//defining motors,servo,sensor
NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE);

AF_DCMotor motor1(2, MOTOR12_8KHZ);
AF_DCMotor motor2(1, MOTOR12_8KHZ);

Servo myservo;

//defining global variables
boolean goesForward=false;

int distance = 100;

int speedSet = 0;

void setup() {

  Serial.begin(9600);

  myservo.attach(10);

  myservo.write(90);

  delay(2000);

  distance = readPing();

  delay(100);

  distance = readPing();

  delay(100);

  distance = readPing();

  delay(100);

  distance = readPing();

  delay(100);

}

void loop() {

  int distanceR = 0;

  int distanceL = 0;

  delay(40);

  Serial.println(distance);

  if(distance<=15)

  {

    Serial.println("Object Detected");

    moveStop();

    delay(100);
```

moveBackward();	{	myservo.write(180);
delay(300);	turnLeft();	
moveStop();	moveStop();	delay(500);
delay(200);	}	int distance =
distanceR =	}else	readPing();
lookRight();	{	delay(100);
	moveForward();	myservo.write(90);
Serial.print("Distance Right = ");	} //reseting the variable after the operations	return distance;
	distance = readPing();	delay(100);
Serial.println(distanceR);	}int lookRight()	}
delay(200);	{	int readPing() {
distanceL = lookLeft();		delay(70);
	myservo.write(0);	int cm = sonar.ping_cm();
Serial.print("Distance Left = ");	delay(500);	if(cm==0)
	int distance = readPing();	{
Serial.println(distanceL);	delay(100);	cm = 250;
delay(200);		}
if(distanceR>=distanceL)	myservo.write(90);	return cm;}
{	return distance;	void moveStop()
turnRight();	}int lookLeft()	{
moveStop();	{	motor1.run(RELEASE);
}else		

motor2.run(RELEASE);	delay(5);	}
void	}	} void
moveForward() {	}	turnRight() {
if(!goesForward)	}	Serial.println("Turning Right");
{	void	motor1.run(FORWARD);
	moveBackward()	
	{	motor2.run(BACKWARD);
goesForward=true;	goesForward=false;	
motor1.run(FORWARD);	motor1.run(BACKWARD);	delay(turn_amount);
motor2.run(FORWARD);	motor2.run(BACKWARD);	motor1.run(FORWARD);
for (speedSet = 0; speedSet < MAX_SPEED; speedSet +=2) // slowly bring the speed up to avoid loading down the batteries too quickly	for (speedSet = 0; speedSet < MAX_SPEED; speedSet +=2) // slowly bring the speed up to avoid loading down the batteries too quickly	motor2.run(FORWARD);
{		} void turnLeft()
		{
motor1.setSpeed(speedSet);	{motor1.setSpeed(speedSet);	Serial.println("Turning Left");
motor2.setSpeed(speedSet+MAX_SPEED_OFFSET);	motor2.setSpeed(speedSet+MAX_SPEED_OFFSET);	motor1.run(BACKWARD);
	delay(5);	motor2.run(FORWARD);

```
delay(turn_amount);  
  
motor2.run(FORWARD);  
  
}  
  
motor1.run(FORWARD);
```

Chapter 5: Results and Discussion

Simulation Output

The simulation of the obstacle-avoiding car was successfully carried out in Arduino IDE. The ultrasonic sensor accurately measured distances, the servo motor scanned the environment, and the motor driver shield controlled the motors effectively. The decision-making algorithm allowed the car to detect obstacles and navigate around them smoothly.

Actual Output

The physical car successfully detected and avoided obstacles in real-world tests. The ultrasonic sensor provided reliable readings, and the servo motor scanned effectively to identify the best path. The motor driver shield accurately controlled the motors, allowing the car to steer away from obstacles without any issues.

Discussion

The project demonstrated efficient obstacle detection and avoidance. The algorithm worked well, but improvements could include:

- Adding more sensors for better coverage.
- Enhancing the decision-making algorithm for faster response.
- Implementing dynamic speed control based on obstacle proximity.

Overall, the obstacle-avoiding car performed well, achieving the desired objectives.

References

1. **Arduino Documentation:** <https://www.arduino.cc/reference/en/>
 2. **HC-SR04 Datasheet:** <https://www.electroschematics.com/hc-sr04-datasheet/>
 3. **Motor Driver Shield Information:** <https://www.adafruit.com/product/1438>
 - 4.
 5. **Servo Motor Datasheet:**
<https://www.towerpro.com.tw/product/sg90-9g-micro-servo/>
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